



Behavioral Proxy Conditioning for Financial Stress Scenario Generation with a Pretrained Diffusion Model

A five-number, human-readable market-state vector turns a pretrained time-series diffusion model into a **controllable calm–stress scenario generator** — no architecture changes, no large financial dataset.

Elena Kuular · Junsuk Choe

Dept. of Computer Science & Engineering, Sogang University · eakuular@sogang.ac.kr

27/28

regime-separation checks passed in every one of 4 seeds (7/7 volatility + 20/21 risk/tail)

2.47±0.08×

stress-to-calm volatility ratio — conditioning alone switches the generated regime

0%

exact replay of history — 1-NN distance to training data below real-window spacing

−7.02%

stress − calm 36-day CVaR₉₅ gap, equal-weight 7-index portfolio in return units

01 · PROBLEM

Crises are rare — generators learn calm markets

History offers a handful of crises drowned in calm data. A generator trained on it looks realistic yet **fails to reproduce the stress regime** risk teams actually need. A valid stress scenario must be, simultaneously:

R Realistic
preserves stylized facts — heavy tails, volatility clustering, leverage effect

C Controllable
calm vs. stress separate measurably in volatility & tail risk, on request

N Novel
not a replay of history — replay cannot reveal unseen risk

02 · KEY IDEA

One interpretable 5-D vector tells the model which market to generate

Each 36-day window gets a **behavioral proxy** — compact, market-observable, comparable across indices. Conditioning on it (plus a market-identity token) lets a **single model** generate calm *and* stress scenarios for all seven indices.

$$\mathbf{C}_i = \left[\begin{array}{|c|c|c|} \hline \mathbf{p}_{\text{calm}} & \mathbf{p}_{\text{normal}} & \mathbf{p}_{\text{stress}} \\ \hline \end{array} \right] \left[\begin{array}{|c|} \hline \mathbf{p}_i \\ \hline \end{array} \right] \left[\begin{array}{|c|} \hline \mathbf{v}_i \\ \hline \end{array} \right]$$

HMM regime probabilities

3-state Gaussian HMM on rolling volatility, fit on train only. **Soft** probabilities capture windows that are partly calm, partly stressed.

Cross-market corr.

Mean pairwise correlation — local vs. systemic shocks.

Volatility rank

Per-index volatility percentile — a scale-free severity indicator.

Proxy-group dropout — robust conditioning

$p = 0.3$ masks one proxy group per draw (weights $[1, 1, 2]$ — the volatility-rank component is dropped twice as often), so the model learns to use **all** components.

Kurtosis-aware fine-tuning — correct tail shape

$\mathcal{L} = \lambda \cdot \max(0, \text{kurt}_{\text{calm}} - \text{kurt}_{\text{stress}} + m)$ penalizes stress samples with lighter tails than calm ones; pooled regime-level directions stabilize it.

03 · METHOD

Pipeline — adapt a pretrained diffusion model, don't build one

1 · DATA

7 global indices, 25 years

SPX FTSE GDAXI N225 SENSEX FCHI AEX

- Daily OHLCV, 2000–2024 · 70/15/15 chronological split
- Log-returns → 36-day windows, stride 9 · z-score normalization
- LVN off — per-window scaling erases the regime signal

2 · CONDITION

Behavioral proxy + market token

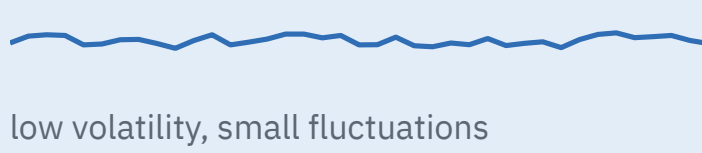
- 5-D proxy per window, globally normalized with train statistics
- Proxy-group dropout ($p = 0.3$) during training
- Market-identity token → one model shares structure across all 7 indices yet keeps index-specific regimes

3 · GENERATION

Fine-tuned ImagenFew + CFG

- ImagenFew — pretrained on 19 time-series datasets · 500 ep. fine-tune → 150 ep. kurtosis FT
- CFG $\hat{y} = \hat{y}_u + s \cdot (\hat{y}_c - \hat{y}_u)$, $s = 1.6$ (validation) → 500 samples / regime / index

CALM · conditioned



low volatility, small fluctuations

STRESS · conditioned

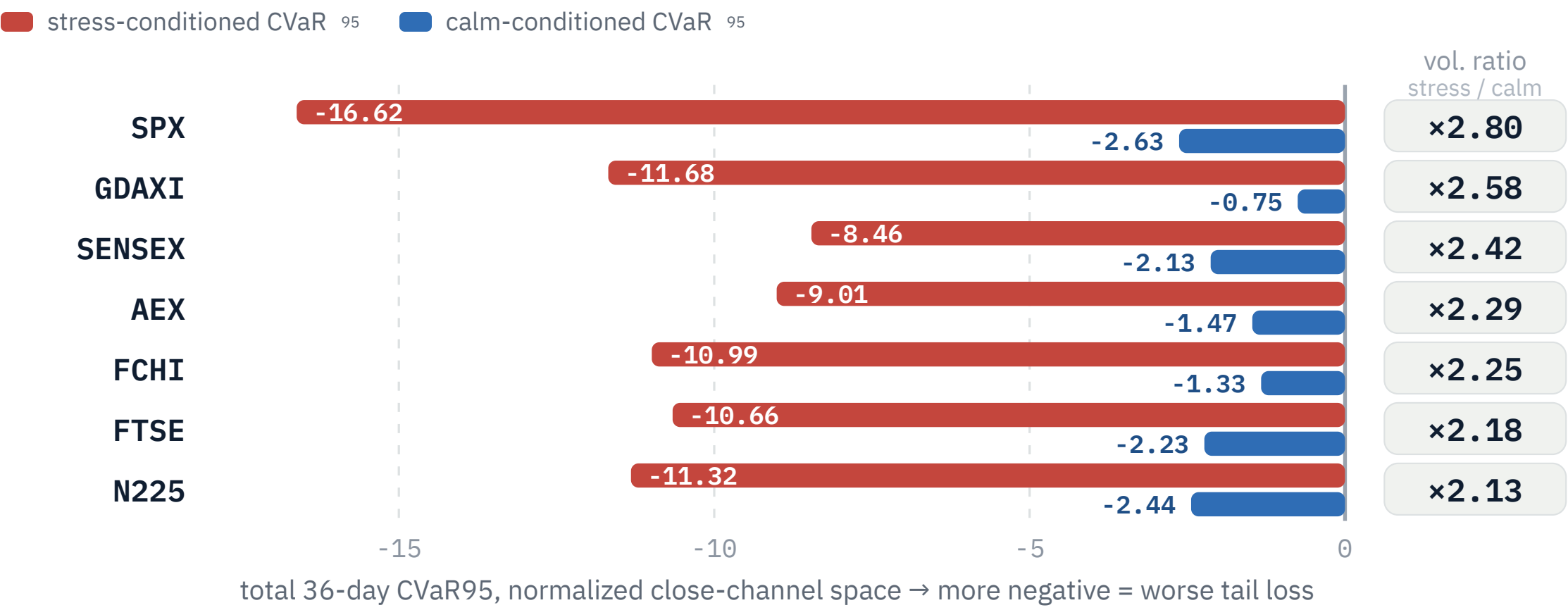


high volatility, heavy tails, drawdowns

04 · DOES THE CONDITIONING WORK?

Clear calm–stress separation across all indices

Stress-conditioned windows show far more adverse tail losses than calm-conditioned ones on all seven markets (*seed 0; every seed passes 27/28*).

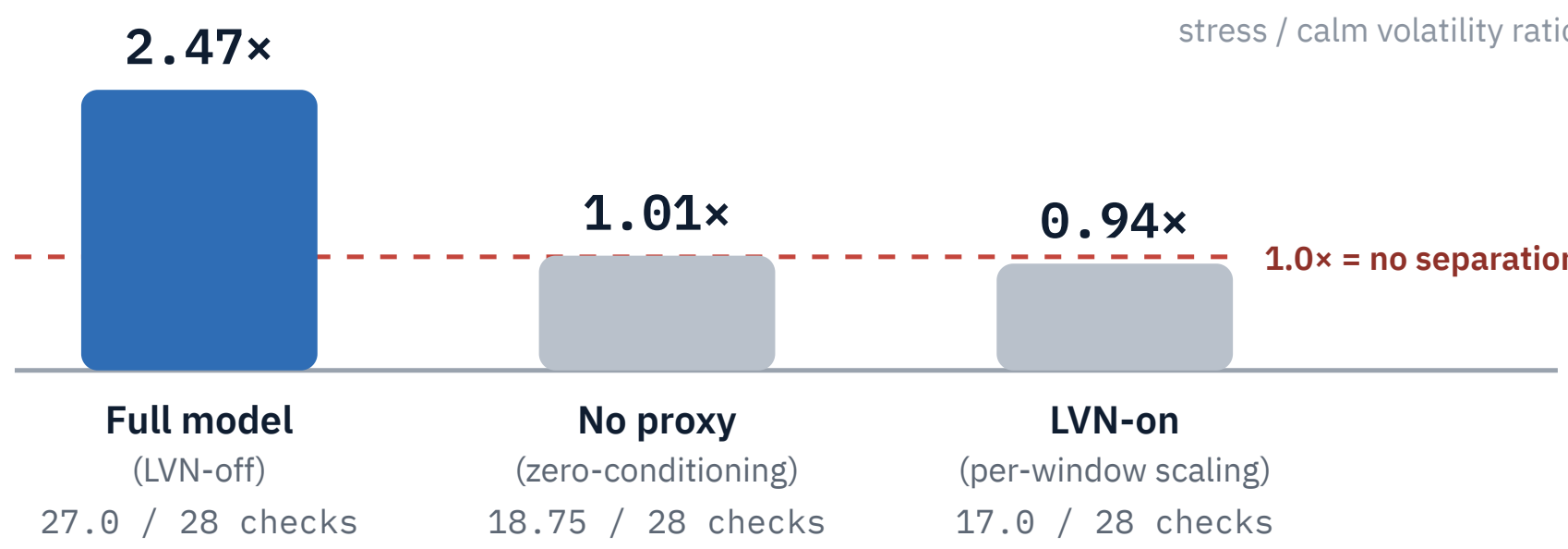


The only failed check is the **GDAXI kurtosis ordering**, while all **14/14 CVaR tail checks** — untied to the conditioning axis — pass.

05 · ABLATIONS

Behavioral proxy controls the regime

Same backbone, no proxy → volatility ratio collapses to **≈1.0×**: no regimes. Per-window volatility scaling (LVN-on) erases it too.



Component ablation (mean checks of 28): base fine-tune **23.3** → + proxy-group dropout **23.8** → + kurtosis FT: **25.8** with window-level directions vs. **27.0** with pooled regime-level directions — the pooled tail-direction estimator lands 27/28 in **every** seed.

06 · REALISM HOLDS

Stylized facts survive the added control

Leverage effect 56/56

OHLCV cross-channel structure 56/56

Non-Gaussianity 55/56

Volatility clustering 43/56

Return ACF 1/56 — open limitation

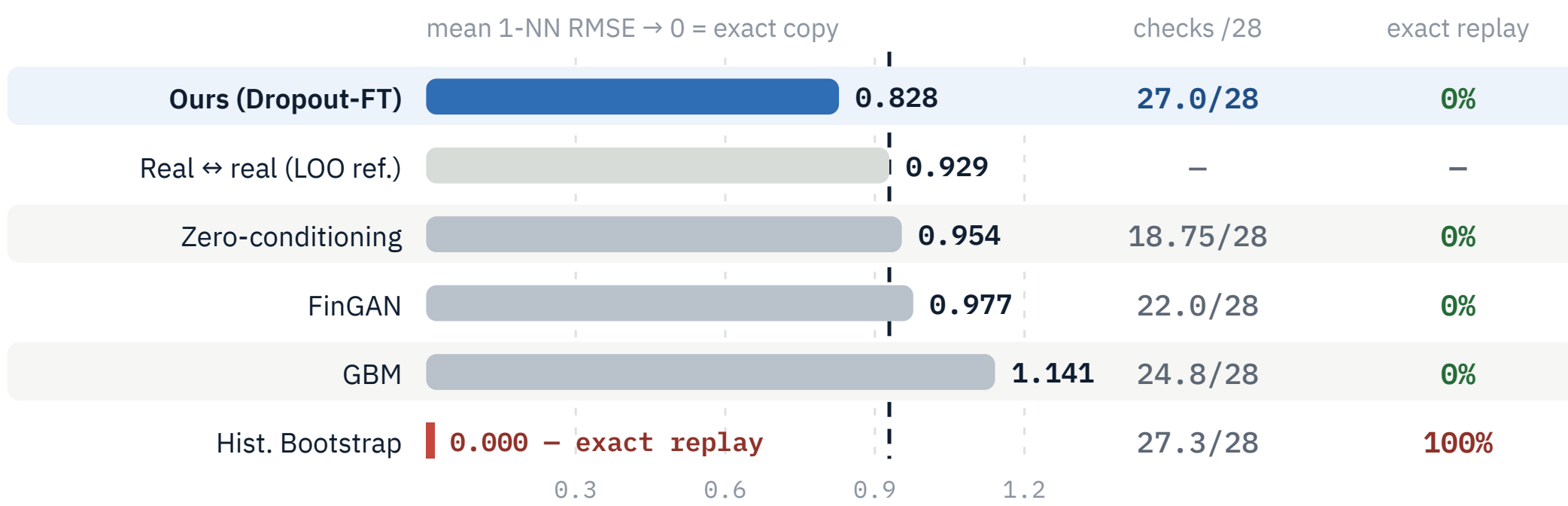
Discriminative score ↓ 0.223 ± 0.003

Context-FID ↓ 2.069 ± 0.076 per fact: 7 indices × 2 regimes × 4 seeds = 56 tests

07 · NOVELTY VS. BASELINES

Separation without replay

Under the same 28-check protocol, our model matches Historical Bootstrap's separation while generating **genuinely new** windows — mean 1-NN RMSE to the training set stays **below real-window spacing without copying**:



Dashed line: leave-one-out spacing **between real windows (0.929)** — generated samples sit inside the real-data manifold; Bootstrap's near-perfect checks come from distance **0.000**, i.e. pure replay.

08 · DOWNSTREAM · HELD-OUT REAL RETURNS

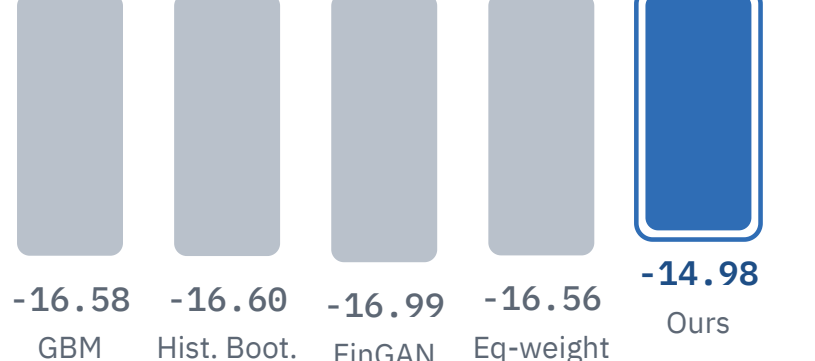
Scenario-driven portfolios cut the deepest losses

Generated calm/stress sets feed a long-only stress-aware optimizer (regime-specific Iman–Conover coupling; 80/20 blend with equal weight); the weights are then evaluated on **real held-out test returns**.

our scenarios GBM · Hist. Bootstrap · FinGAN · Equal-weight

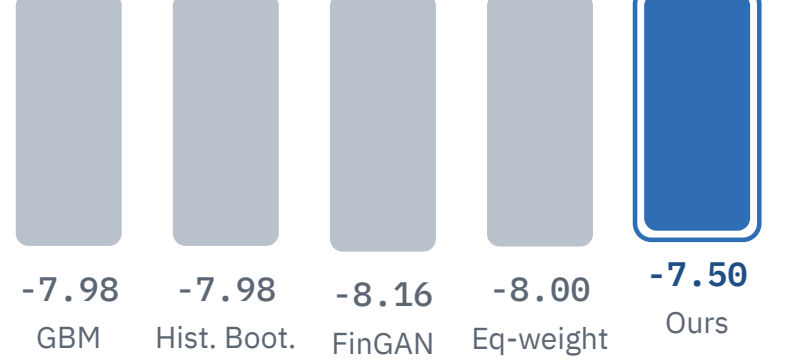
Maximum drawdown

% — closer to 0 is better



36-day CVaR₉₅

% — closer to 0 is better



Consistently best on the reported downside-risk metrics — Calmar **0.60** (vs. 0.54–0.55), daily CVaR₉₅ **−1.73%** — returns on par (8.98%): the gain is **loss control, not higher returns**.

One realized test path; stationary-bootstrap CIs are wide → **consistent directional evidence**, not certified outperformance.

Takeaways

→ Controllability comes from the proxy, not the backbone

An interpretable 5-D behavioral vector is enough: removing it collapses the stress/calm volatility ratio from **2.47×** to **≈1.0×** on the identical pretrained model.

→ Novel scenarios, not replayed data

27/28 checks with 0% replay, matching a replay-based baseline while generating novel scenarios. Separation survives conversion to return units (≈2.3× vol, −7.02% CVaR₉₅ gap).

→ The learned stress signal improves portfolio allocation

Portfolios optimized on generated scenarios show the **shallowest drawdowns** on unseen real markets — the learned regimes carry genuine risk information.

Limitations & future work. Volatility rank contributes to both the proxy construction and the evaluation split — volatility checks are controllability evidence, CVaR checks the independent ones · cross-asset dependence is corrected post-hoc, not learned · return ACF remains open. Next: joint multivariate generation and learned cross-asset dependence; broader asset classes and longer scenario horizons.